

RESEARCH DATA TEMPLATE — DATA CONTRACT

Claude Code: Populate every RESEARCH_REQUIRED field with live web research | Save as output/research_data.json

INSTRUCTIONS: This document defines every data field the Excel model requires. For each stock, gather all fields marked RESEARCH_REQUIRED via web search. NVO is always the first stock and requires the deep_dive section. Add 4–6 additional stocks using the same schema. Remove all _INSTRUCTIONS fields before saving the final JSON.

METADATA

data_sources

[1]

[2]

[3]

as_of_date	RESEARCH_REQUIRED: today's date YYYY-MM-DD
model_version	1.0

STOCKS — DATA FIELDS PER STOCK (NVO shown as primary example)

Stock 1: NVO

NOTE: NVO is ALWAYS first. Populate all fields with current data.

scores

valuation

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: 1-2 sentences with specific data (P/E, EV/EBITDA vs. history)

balance_sheet

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: 1-2 sentences with net debt, coverage, credit rating

catalyst

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: name the top 1-2 catalysts and timeline

selloff_quality

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: explain what drove the selloff and whether it's overblown

■ industry_protection

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: how recession-proof is GLP-1/diabetes demand?

■ analyst_conviction

score	RESEARCH_REQUIRED: 0-10
justification	RESEARCH_REQUIRED: describe current analyst consensus and recent moves

■ justifications

valuation	RESEARCH_REQUIRED
balance_sheet	RESEARCH_REQUIRED
catalyst	RESEARCH_REQUIRED
selloff_quality	RESEARCH_REQUIRED
industry_protection	RESEARCH_REQUIRED
analyst_conviction	RESEARCH_REQUIRED

■ valuation

fwd_pe	RESEARCH_REQUIRED
sector_avg_pe	RESEARCH_REQUIRED: large pharma sector average forward P/E
hist_5yr_pe	RESEARCH_REQUIRED: NVO's own 5-year avg P/E
ev_ebitda	RESEARCH_REQUIRED
peer_avg_ev_ebitda	RESEARCH_REQUIRED: LLY, ABBV, MRK, PFE average
p_fcf	RESEARCH_REQUIRED
ev_sales	RESEARCH_REQUIRED
bear_target	RESEARCH_REQUIRED
base_target	RESEARCH_REQUIRED
bull_target	RESEARCH_REQUIRED
analyst_avg_target	RESEARCH_REQUIRED
dcf_notes	RESEARCH_REQUIRED: brief note on DCF methodology used

■ fundamentals

■ balance_sheet

net_cash_debt_b	RESEARCH_REQUIRED: positive = net cash, negative = net debt
nd_ebitda	RESEARCH_REQUIRED
interest_coverage	RESEARCH_REQUIRED
credit_rating	RESEARCH_REQUIRED

■ cash_flow

fcf_ltm_b	RESEARCH_REQUIRED
capex_rev_pct	RESEARCH_REQUIRED: decimal
market_cap_b	RESEARCH_REQUIRED
ev_b	RESEARCH_REQUIRED

rev_ltm_b	RESEARCH_REQUIRED: in USD billions
rev_ntm_b	RESEARCH_REQUIRED: consensus estimate
ebitda_margin	RESEARCH_REQUIRED: decimal
eps_ltm	RESEARCH_REQUIRED: USD, ADR basis
eps_ntm	RESEARCH_REQUIRED: consensus estimate
eps_3yr_cagr	RESEARCH_REQUIRED: decimal
div_yield	RESEARCH_REQUIRED: decimal

■ catalysts

[1]

description	RESEARCH_REQUIRED: e.g. Zenagamtide Phase 3 data readout
timeline	RESEARCH_REQUIRED: e.g. Q3 2026
probability_decimal	RESEARCH_REQUIRED: e.g. 0.45
impact_if_hit_pct	RESEARCH_REQUIRED: e.g. 25 for +25%
impact_if_miss_pct	RESEARCH_REQUIRED: e.g. -10 for -10%
monitor_signal	RESEARCH_REQUIRED: what to watch for

[2]

description	RESEARCH_REQUIRED
timeline	RESEARCH_REQUIRED
probability_decimal	RESEARCH_REQUIRED
impact_if_hit_pct	RESEARCH_REQUIRED
impact_if_miss_pct	RESEARCH_REQUIRED
monitor_signal	RESEARCH_REQUIRED

[3]

description	RESEARCH_REQUIRED
timeline	RESEARCH_REQUIRED
probability_decimal	RESEARCH_REQUIRED
impact_if_hit_pct	RESEARCH_REQUIRED
impact_if_miss_pct	RESEARCH_REQUIRED
monitor_signal	RESEARCH_REQUIRED

■ scenarios

■ bull

description	RESEARCH_REQUIRED: what has to go right (specific, not generic)
price_target	RESEARCH_REQUIRED
return_pct	RESEARCH_REQUIRED: decimal
probability	RESEARCH_REQUIRED: decimal, e.g. 0.25

■ base

description	RESEARCH_REQUIRED
price_target	RESEARCH_REQUIRED

return_pct	RESEARCH_REQUIRED: decimal
probability	RESEARCH_REQUIRED: decimal, e.g. 0.50

■ bear

description	RESEARCH_REQUIRED: what has to go wrong
price_target	RESEARCH_REQUIRED
return_pct	RESEARCH_REQUIRED: decimal, negative number
probability	RESEARCH_REQUIRED: decimal, e.g. 0.25

■ deep_dive

■ selloff_timeline

[1]

date	RESEARCH_REQUIRED: e.g. Jun 2024
event	RESEARCH_REQUIRED
price	RESEARCH_REQUIRED
pct_change	RESEARCH_REQUIRED: decimal
interpretation	RESEARCH_REQUIRED

■ pipeline

[1]

asset	Wegovy (Oral)
indication	Obesity
phase	Approved/Launched
status	RESEARCH_REQUIRED
timeline	Launched 2025
market_size_b	RESEARCH_REQUIRED
probability	0.9
bear_rev_m	RESEARCH_REQUIRED
base_rev_m	RESEARCH_REQUIRED
bull_rev_m	RESEARCH_REQUIRED

[2]

asset	CagriSema
indication	Obesity / T2D
phase	Filed (FDA decision late 2026)
status	RESEARCH_REQUIRED
timeline	FDA decision H2 2026
market_size_b	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED
bear_rev_m	RESEARCH_REQUIRED
base_rev_m	RESEARCH_REQUIRED
bull_rev_m	RESEARCH_REQUIRED

[3]

asset	Zenagamtide (Amycretin)
indication	Obesity
phase	Phase 3 (planned H2 2026 start)
status	RESEARCH_REQUIRED
timeline	Data: 2027-2028
market_size_b	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED
bear_rev_m	RESEARCH_REQUIRED
base_rev_m	RESEARCH_REQUIRED
bull_rev_m	RESEARCH_REQUIRED

[4]

asset	Semaglutide (CKD / HFpEF labels)
indication	Chronic Kidney Disease / Heart Failure
phase	Label expansion
status	RESEARCH_REQUIRED
timeline	RESEARCH_REQUIRED
market_size_b	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED
bear_rev_m	RESEARCH_REQUIRED
base_rev_m	RESEARCH_REQUIRED
bull_rev_m	RESEARCH_REQUIRED

[5]

asset	MASH Pipeline (Akeru acq.)
indication	MASH / Metabolic Liver Disease
phase	Early-Stage / Phase 2
status	RESEARCH_REQUIRED
timeline	RESEARCH_REQUIRED
market_size_b	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED
bear_rev_m	RESEARCH_REQUIRED
base_rev_m	RESEARCH_REQUIRED
bull_rev_m	RESEARCH_REQUIRED

■ ira_impact

current_net_price	RESEARCH_REQUIRED: current US net price per month
ira_ozempic_price	274
ira_wegovy_price	385
revenue_at_risk_b	RESEARCH_REQUIRED
pct_revenue_at_risk	RESEARCH_REQUIRED: decimal

eps_impact_bear	RESEARCH_REQUIRED: USD per ADR share
eps_impact_base	RESEARCH_REQUIRED: USD per ADR share

■ competitive_positioning

nvo_glp1_market_share_pct	RESEARCH_REQUIRED
lly_glp1_market_share_pct	RESEARCH_REQUIRED
trend	RESEARCH_REQUIRED: gaining / losing / stable
notes	RESEARCH_REQUIRED

■ ma_capacity

available_cash_b	RESEARCH_REQUIRED
leverage_headroom_b	RESEARCH_REQUIRED: how much more debt at 3x ND/EBITDA
total_firepower_b	RESEARCH_REQUIRED
likely_targets	RESEARCH_REQUIRED

■ verdict

conclusion	RESEARCH_REQUIRED: one of: YES — Priced In / PARTIALLY — Still Risks Ahead / NO — More Pain Likely
bullish	RESEARCH_REQUIRED: true or false
rationale	RESEARCH_REQUIRED: 3-5 sentence paragraph at sell-side analyst quality. Be specific: cite price, multiples, pipeline, IRA risk, competitive dynamics. Provide a clear investment recommendation with a target price and time horizon.

ticker	NVO
company	Novo Nordisk A/S
industry	Pharmaceuticals — GLP-1/Diabetes/Obesity
current_price	RESEARCH_REQUIRED
week52_high	RESEARCH_REQUIRED
week52_low	RESEARCH_REQUIRED
pct_off_high	RESEARCH_REQUIRED: decimal e.g. -0.73 for -73%
composite_score	RESEARCH_REQUIRED: 0-10
recommendation	RESEARCH_REQUIRED: STRONG BUY / BUY / SPECULATIVE BUY / PASS
target_price	RESEARCH_REQUIRED: your base case target
implied_upside	RESEARCH_REQUIRED: decimal
expected_value_pct	RESEARCH_REQUIRED: probability-weighted return, decimal
key_risk	RESEARCH_REQUIRED: single most important risk, one sentence
key_catalyst	RESEARCH_REQUIRED: single most important catalyst, one sentence
analyst_consensus	RESEARCH_REQUIRED: e.g. Hold (6 Buy / 10 Hold / 0 Sell)

Stock 2: RESEARCH_REQUIRED: your chosen ticker

■ NOTE: Stock #2 — Claude Code finds and populates. Must pass all hard filters. Copy/extend this template for each additional stock. Suggest evaluating: MRK, BMY, PFE, BSX, ZBH, ABT, LMT, RTX, GD, ABBV

■ scores

■ valuation

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ balance_sheet

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ catalyst

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ selloff_quality

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ industry_protection

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ analyst_conviction

score	RESEARCH_REQUIRED
justification	RESEARCH_REQUIRED

■ justifications

valuation	RESEARCH_REQUIRED
balance_sheet	RESEARCH_REQUIRED
catalyst	RESEARCH_REQUIRED
selloff_quality	RESEARCH_REQUIRED
industry_protection	RESEARCH_REQUIRED
analyst_conviction	RESEARCH_REQUIRED

■ valuation

fwd_pe	RESEARCH_REQUIRED
sector_avg_pe	RESEARCH_REQUIRED
hist_5yr_pe	RESEARCH_REQUIRED
ev_ebitda	RESEARCH_REQUIRED
peer_avg_ev_ebitda	RESEARCH_REQUIRED
p_fcf	RESEARCH_REQUIRED
ev_sales	RESEARCH_REQUIRED
bear_target	RESEARCH_REQUIRED
base_target	RESEARCH_REQUIRED
bull_target	RESEARCH_REQUIRED

analyst_avg_target	RESEARCH_REQUIRED
dcf_notes	RESEARCH_REQUIRED

■ fundamentals

■ balance_sheet

net_cash_debt_b	RESEARCH_REQUIRED
nd_ebitda	RESEARCH_REQUIRED
interest_coverage	RESEARCH_REQUIRED
credit_rating	RESEARCH_REQUIRED

■ cash_flow

fcf_ltm_b	RESEARCH_REQUIRED
capex_rev_pct	RESEARCH_REQUIRED
market_cap_b	RESEARCH_REQUIRED
ev_b	RESEARCH_REQUIRED
rev_ltm_b	RESEARCH_REQUIRED
rev_ntm_b	RESEARCH_REQUIRED
ebitda_margin	RESEARCH_REQUIRED
eps_ltm	RESEARCH_REQUIRED
eps_ntm	RESEARCH_REQUIRED
eps_3yr_cagr	RESEARCH_REQUIRED
div_yield	RESEARCH_REQUIRED

■ catalysts

[1]

description	RESEARCH_REQUIRED
timeline	RESEARCH_REQUIRED
probability_decimal	RESEARCH_REQUIRED
impact_if_hit_pct	RESEARCH_REQUIRED
impact_if_miss_pct	RESEARCH_REQUIRED
monitor_signal	RESEARCH_REQUIRED

■ scenarios

■ bull

description	RESEARCH_REQUIRED
price_target	RESEARCH_REQUIRED
return_pct	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED

■ base

description	RESEARCH_REQUIRED
price_target	RESEARCH_REQUIRED

return_pct	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED

■ bear

description	RESEARCH_REQUIRED
price_target	RESEARCH_REQUIRED
return_pct	RESEARCH_REQUIRED
probability	RESEARCH_REQUIRED

ticker	RESEARCH_REQUIRED: your chosen ticker
company	RESEARCH_REQUIRED
industry	RESEARCH_REQUIRED
current_price	RESEARCH_REQUIRED
week52_high	RESEARCH_REQUIRED
week52_low	RESEARCH_REQUIRED
pct_off_high	RESEARCH_REQUIRED
composite_score	RESEARCH_REQUIRED
recommendation	RESEARCH_REQUIRED
target_price	RESEARCH_REQUIRED
implied_upside	RESEARCH_REQUIRED
expected_value_pct	RESEARCH_REQUIRED
key_risk	RESEARCH_REQUIRED
key_catalyst	RESEARCH_REQUIRED
analyst_consensus	RESEARCH_REQUIRED

■ SCREENED OUT — STOCKS THAT FAILED HARD FILTERS

[1]

■ NOTE: Add any stocks Claude Code considered but rejected. Be specific about the filter they failed.

ticker	RESEARCH_REQUIRED
company	RESEARCH_REQUIRED
filter_failed	RESEARCH_REQUIRED: e.g. Balance Sheet, Industry Protection
reason	RESEARCH_REQUIRED
revisit_if	RESEARCH_REQUIRED: what would have to change to reconsider

■ ASSUMPTIONS SCHEMA

■ macro

[1]

label	Risk-Free Rate (10-Yr Treasury)
value	RESEARCH_REQUIRED: current yield

notes	Used as discount rate floor
source	US Treasury website

[2]

label	Equity Risk Premium
value	5.5%
notes	Damodaran US ERP estimate
source	NYU Damodaran

[3]

label	Pharma Sector WACC
value	8.0%
notes	Used for DCF unless company-specific data available
source	Industry standard

[4]

label	Terminal Growth Rate
value	2.5%
notes	Conservative; below long-run nominal GDP
source	Analyst judgment

[5]

label	US GDP Growth Forecast 2026
value	RESEARCH_REQUIRED
notes	Bearish macro assumption
source	RESEARCH_REQUIRED

[6]

label	US Credit Conditions
value	Tightening
notes	Investor macro thesis — bearish on credit
source	Investor mandate

■ valuation

[1]

label	DCF Weight in Blended Target
value	33%
notes	Equal weight with comps and analyst target
source	Methodology

[2]

label	Comps Weight in Blended Target
value	33%
notes	
source	Methodology

[3]

label	Analyst Consensus Weight
value	33%
notes	
source	Methodology

[4]

label	Holding Period
value	12–18 months
notes	Tactical horizon for this strategy
source	Investor mandate

[5]

label	Target Return
value	20–50%
notes	Minimum acceptable risk/reward
source	Investor mandate

[6]

label	Max Acceptable Drawdown
value	15–20%
notes	Stop-loss level
source	Investor mandate

■ peer_groups

[1]

ticker	NVO
peers	LLY, ABBV, MRK, AZN, RHHBY
rationale	Global large-cap pharma with diabetes/obesity or primary care focus
exclusions	Pure specialty pharma, biotech under \$20B market cap

■ RESEARCH_REQUIRED

Field must be populated with live web research before running build_model.py

■ Pre-filled value

Default or known value — verify and update if needed

■ NOTE box

Instructions from the schema — remove from final JSON